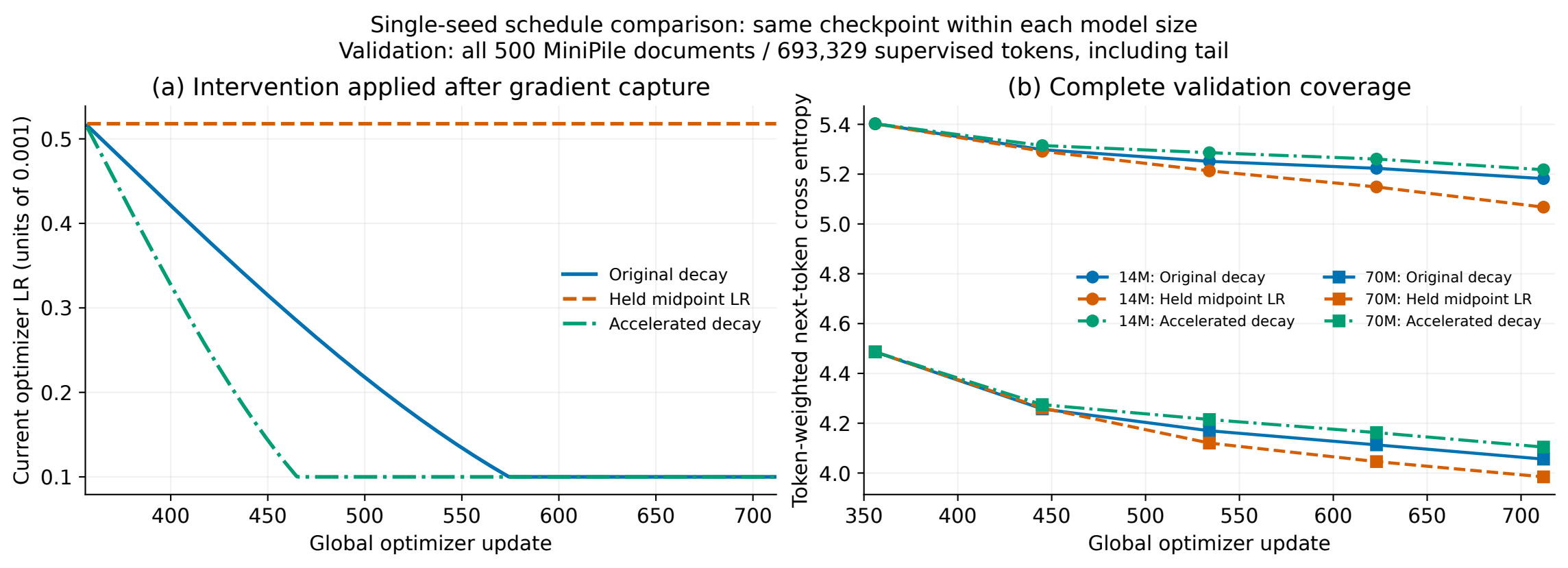
Single-seed schedule comparison: same checkpoint within each model size
Validation: all 500 MiniPile documents / 693,329 supervised tokens, including tail
(a) Intervention applied after gradient capture
Current optimizer LR (units of 0.001)
0.5
0.4
0.3
0.2
0.1
400
450
500
550
600
650
700
Global optimizer update
Original decay
Held midpoint LR
Accelerated decay
(b) Complete validation coverage
Token-weighted next-token cross entropy
5.4
5.2
5.0
4.8
4.6
4.4
4.2
4.0
350
400
450
500
550
600
650
700
Global optimizer update
14M: Original decay
14M: Held midpoint LR
14M: Accelerated decay
70M: Original decay
70M: Held midpoint LR
70M: Accelerated decay